

Advanced Lesson 1 - Exercises: The Mathematica Interface

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MATH 7013 - Advanced Mathematical Modeling

1: Front-end tour - toolbar, magnification, Documentation Center

This exercise is entirely a walk through the notebook front end: show the toolbar, raise the magnification, open the Documentation Center, look up Integrate, and read the seven linked tutorials (Your First Mathematica Calculations, Arithmetic, Defining Variables, Defining Functions, Some Mathematical Functions, Symbolic Calculations, Numerical Calculations). The professor's own instruction is "Don't worry about an answer--just do it!" There is nothing to evaluate and nothing to submit.

One thing worth doing while the Documentation Center is open: the Basic Math Input palette (Palettes menu) supplies the typeset forms this lesson keeps asking for - the superscript template, the partial-derivative template, the integral-with-differential template, and the Pi, E and I characters. Every one of those has a keyboard equivalent, and the exercises below are written in ordinary linear syntax, which evaluates identically.

2: The numerical function N

$$N\left[\frac{1}{13}\right]$$

$$N\left[\frac{1}{13}, 50\right]$$

N with one argument returns a machine-precision real: about sixteen digits, carried in hardware. N with a second argument asks for that many digits of precision and Mathematica switches to software arbitrary-precision arithmetic to supply them.

3: Assignment, and what "?" reports

```
Clear[a, A];
```

```
a A
```

```
?a
```

```
?A
```

```
a = 5
```

```
a A
```

```
?a
```

```
?A
```

```
A = 6
```

```
a A
```

```
?a
```

```
?A
```

```
A = 7
```

```
a A
```

```
?a
```

```
?A
```

Results. With nothing assigned, aA stays as the symbolic product aA , and "?" reports each symbol with no value attached. After $a = 5$ the product becomes $5A$: Mathematica substitutes every value it has and leaves the rest symbolic. After $A = 6$ the product is 30 , and after $A = 7$ it is 35 - the new assignment to A simply overwrites the old one. Note that a and A are different symbols; Mathematica is case sensitive, which is exactly why the convention is that built-in names start with a capital and yours do not.

4: Sin versus sin

```
?Sin
```

```
?sin
```

"?Sin" reports the built-in: $\text{Sin}[z]$ gives the sine of z , with a link into the Documentation Center. "?sin" reports a bare symbol with no value, no

definition and no usage message, because no such built-in exists. Lowercase `sin` is just an ordinary undefined symbol - which is precisely why user functions are given lowercase names: there is no chance of colliding with a protected built-in.

5: Symbols as units

```
Clear[a, A, feet, sec];
```

$$\frac{a}{A}$$

$$\frac{a}{A}$$

```
?a
```

```
?A
```

$$A = \frac{5 \text{ feet}}{\text{sec}}$$

$$\frac{a}{A}$$

$$\frac{a}{A}$$

```
a = 10 feet
```

$$\frac{a}{A}$$

$$\frac{a}{A}$$

```
?feet
```

```
?sec
```

Results. With nothing assigned, a/A is the symbolic quotient. After $A = 5 \text{ feet/sec}$ the quotient becomes $a \text{ sec} / (5 \text{ feet})$. After $a = 10 \text{ feet}$ the quotient becomes 2 sec : the symbol "feet" cancels between numerator and denominator by ordinary algebra, leaving a pure time. "`?feet`" and "`?sec`" report symbols with no values - they never needed any. That is the whole trick. Undefined symbols behave as formal algebraic units and cancel correctly without any units package at all.

6: Defining a function, differentiating, integrating, solving, plotting

```
Clear[x, f];
```

```
f[x_] = Sin[x];
```

$$\{f[\pi], f\left[\frac{\pi}{2}\right], f[6.]\}$$

$f[\text{Pi}]$ returns exactly 0 and $f[\text{Pi}/2]$ returns exactly 1, because Pi is an exact symbolic constant and Mathematica knows those two values. $f[6.0]$ returns an approximate real, because the argument was approximate: an inexact input forces an inexact output.

?f

$\partial_x f[x]$

$f'[x]$

$\int f[x] \, dx$

`Solve[3 f[x] + 4 == 0, x]`

`Plot[f[x], {x, 0, 2 Pi}]`

Both derivative notations return $\text{Cos}[x]$; $f'[x]$ is shorthand for the same Derivative operation. The integral is $-\text{Cos}[x]$, with no constant of integration - Mathematica returns one antiderivative, not the family. Solve does not return a single principal value. It returns two complete solution families, each a ConditionalExpression carrying the side condition that the arbitrary integer $C[1]$ is an integer: $x = -\text{ArcSin}[4/3] + 2 \text{Pi } C[1]$ and $x = \text{Pi} + \text{ArcSin}[4/3] + 2 \text{Pi } C[1]$. That is the honest answer for a periodic equation - infinitely many solutions, parameterised. Note also that $4/3$ lies outside the range of the real sine, so $\text{ArcSin}[4/3]$ is complex and every one of those solutions is a complex number.

7: $f[x] = x^2$ versus $f[x_] = E^x$ versus $f[2] = \text{Pi}$

`Clear[a, x, f, y];`

`f[x] = x^2;`

`f[x_] = e^x;`

`f[2] = Pi;`

`{f[x], f[a], f[y], f[2], f[3]}`

?f

Three separate definitions are now stored, and Mathematica orders them from most specific to most general: $f[2]$ first, then $f[x]$ (a literal argument, not a pattern), then $f[x_]$.

$f[x]$ returns x^2 - the literal definition wins for that one symbol.

$f[a]$ and $f[y]$ return E^a and E^y - no literal definition matches, so the pattern definition applies.

$f[2]$ returns Pi - the literal definition for 2 wins over the pattern.

$f[3]$ returns E^3 - only the pattern matches.

The lesson: $f[x] = x^2$ without the underscore defines the function at the single symbol x and nowhere else. It is almost never what you want.

8: Immediate versus delayed assignment with RandomReal

```
Clear[f1, f2, x];
?RandomReal
f1[x_] = RandomReal[] x;
{f1[x], f1[x], f1[x]}
Plot[f1[x], {x, 0, 1}]
?f1
f2[x_] := RandomReal[] x;
{f2[x], f2[x], f2[x]}
Plot[f2[x], {x, 0, 1}]
?f2
```

Results. $f1$ gives the same coefficient all three times, and its plot is a straight line through the origin: with immediate assignment the right-hand side was evaluated once, at definition time, so `RandomReal` ran once and the number it produced was frozen into the definition. "`?f1`" shows a definite numeric coefficient stored in the rule.

$f2$ gives three different values, and its plot is noise: with delayed assignment the right-hand side is re-evaluated at every call, so `RandomReal` runs afresh for each of the several hundred sample points `Plot` requests. "`?f2`" shows `RandomReal[]*x` stored literally, unevaluated.

9: Derivative, integral, radical equation, and a cubic plot

```
Clear[x, n];
 $\partial_x x^n$ 
 $\int x^n dx$ 
Solve[x +  $\sqrt{x} == 9$ , x]
Plot[x^3 - 2 x + 1, {x, -2, 2}]
```

The derivative is $n x^{(n-1)}$ as expected. The integral is $x^{(n+1)}/(n+1)$, returned without any condition - Mathematica quietly gives the generic

answer and does not stop to warn that $n = -1$ is excluded. The radical equation has the single solution $x = (19 - \text{Sqrt}[37])/2$, and Solve discards the extraneous root that squaring would introduce. The cubic has a local maximum near $x = -0.8$ and a local minimum near $x = 0.8$, with one real root in view to the left.

10: Wildcard searches on the name Style

```
?Style
```

```
Names["Style*"]
```

```
Length[Names["Style*"]]
```

```
Names["*Style"]
```

```
Length[Names["*Style"]]
```

```
Names["*Style*"]
```

```
Length[Names["*Style*"]]
```

The wildcard "*" stands for any run of characters, so "Style*" finds names beginning with Style, "*Style" finds names ending in Style, and "*Style*" finds names containing it anywhere. Names returns the list; Length counts it. In this version of the language the counts are 13, 107 and 137. The exercise quotes "about 12, about 45, about 65", which were the counts when it was written; the built-in name list has grown substantially since, and the "ends in Style" family in particular has more than doubled. The first count is still essentially right because Style* names are core language constructs, while the *Style names are mostly plot and graph options, which is where most of the growth has happened.